



December 2025

(For EC)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

20 Marks

- 1-

12. K-mode clustering is suitable for
 A) Categorical data B) Numerical data
 C) Mixed data D) Binary data
13. The main steps in in Expectation-Maximization algorithm are
 A) Cluster and merge B) Estimate and maximize likelihood
 C) Select and prune D) Encode and decode
14. Gaussian Mixture Models assume that data is generated from
 A) A single Gaussian distribution B) Uniform distributions only
 C) A mixture of multiple Gaussian distributions D) Random noise
15. Principal Component Analysis (PCA) reduces dimensionality by
 A) Minimizing data redundancy using clustering B) Selecting random features
 C) Minimizing entropy D) Maximizing variance captured in fewer dimensions
16. Bayes theorem is used to compute
 A) Posterior probability given prior and likelihood B) Likelihood given posterior and prior
 C) Prior probability given posterior D) Maximum likelihood estimate
17. Least Squares error in hypothesis evaluation refers to
 A) Squared difference between probabilities of two hypotheses B) Squared difference between entropy values
 C) Squared difference between actual and predicted outputs D) Squared difference between prior and posterior probabilities
18. The Naïve Bayes classifier assumes
 A) All attributes are dependent B) Attributes are conditionally independent given the class
 C) Attributes follow Gaussian distribution D) Data has no noise
19. A common application of EM algorithm is
 A) Training decision trees B) Solving linear regression
 C) Feature scaling D) Estimating parameters of Gaussian Mixture Models
20. Naïve Bayes is particularly effective when
 A) Data is high-dimensional B) Features are strongly correlated
 C) There is no missing data D) Classes are linearly separable

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Describe the steps in designing a machine learning system with a suitable example.
 b) Apply Find-S algorithm to derive final hypothesis for the given training dataset.

Outlook	Temp	Humidity	Wind	Play
Sunny	Warm	Normal	Strong	Yes
Sunny	Warm	High	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

2. a) Explain the major issues and perspectives in machine learning?

Marks	BT*	CO*	PO*
8	L*2	1	1
8	L3	2	1
8	L2	1	1

- b) Apply the Candidate Elimination Algorithm to compute the Version Space (S and G boundaries) for the given dataset.

Outlook	Temp	Humidity	Wind	Play
Sunny	Warm	Normal	Strong	Yes
Sunny	Warm	High	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

8 L3 2 1

3. a) Explain the types of Machine Learning with examples.
b) Construct the decision tree using ID3 algorithm using the given dataset.

8 L2 1 1

Outlook	Play tennis
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	Yes

8 L3 2 1

Unit – II

4. a) What are kernel functions in SVM? Describe different types of kernels.
b) Apply k-means clustering for the given data, compute two cluster points with initial cluster centers (1.0, 1.0) and (5.0, 7.0). Execute for two iterations.

8 L2 3 1

Record No.	A	B
R1	1.0	1.0
R2	1.5	2.0
R3	3.0	4.0
R4	5.0	7.0
R5	3.5	5.0
R6	4.5	5.0
R7	3.5	4.5

8 L3 4 1

5. a) Explain different model combination schemes in ensemble learning.
b) Apply Principal Component Analysis for the given dataset to reduce the features from two to one.

8 L2 3 1

Feature	Ex1	Ex2	Ex3	Ex4
X1	4	8	13	7
X2	11	4	5	14

8 L3 4 1

6. a) Explain Error-Correcting Output Codes for multi-class classification.
b) What are Gaussian Mixture Models? How are they different from K-means?

8 L2 3 1

8 L2 4 1

Unit – III

7. a) State Bayes theorem and explain its importance in machine learning. A test for a disease is 90% accurate. The probability that a person has the disease is 5%. If the test result is positive, calculate the probability that the person actually has the disease.

8 L3 5 1

EC3262-1

SEE - December 2025

- b) Explain the Minimum Description Length principle in learning. Hypothesis H1 requires 120 bits, H2 requires 90 bits. Which will MDL select?

8 L3 5 1

8. a) Explain the 'Naïve' assumption in Naïve Bayes.

Use Naïve Bayes to classify an email with word 'FREE' given:
 $P(\text{Spam})=0.5$, $P(\text{Not Spam})=0.5$, $P(\text{Free}|\text{Spam})=0.8$, $P(\text{Free}|\text{Not Spam})=0.2$

8 L3 5 1

- b) Explain the principle of Maximum Likelihood estimation and Least Squares error minimization in the context of machine learning. Consider data points $(x,y)=\{(1,2),(2,3),(3,5)\}$. Assume a linear model $y=wx$ without intercept. Estimate parameter w using Least Squares error minimization.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
